

Worksheet 4C: Advanced Rules — Derivatives and Their Integral Inverses — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. $3x^2 + 2x$
2. $4t^3 - 12t^2$
3. $2p - 3$
4. $12x - 1$
5. $5m^4 + 12m^3 - 3m^2$
6. $4r^3$
7. $2x(x+1)^3 + x^2 \cdot 3(x+1)^2 = x(x+1)^2(5x+2)$
8. $2(t+1)(t-1)^3 + (t+1)^2 \cdot 3(t-1)^2 = (t+1)(t-1)^2(5t-1)$
9. $(2k+1)^4 + k \cdot 4(2k+1)^3 \cdot 2 = (2k+1)^3(10k+1)$
10. $2(n^2+1) \cdot 2n \cdot (n-1) + (n^2+1)^2 = (n^2+1)(5n^2-4n+1)$

Part B

11. $x^3 + x^2 + C$ (or $x^2(x+1) + C$)
12. $x^2 + 2x + C$ (or $(x-1)(x+3) + C = x^2 + 2x - 3 + C$)
13. Derivative: $4x - 3$; integral gives $(2x+1)(x-2) + C$
14. $x^3 + x^2 + C$
15. $x^2 + 2x + C$
16. Yes (differ by constant)

Part C

17. $\frac{2}{(x+2)^2}$
18. $\frac{t^2-2t}{(t-1)^2}$

19. $\frac{-2}{(p-1)^2}$
20. $1 - \frac{1}{x^2}$
21. $\frac{-2m}{(m^2+1)^2}$
22. $\frac{r^4+3r^2}{(r^2+1)^2}$
23. Both: $-3x^{-4}$
24. Simplify to $x^2 + 2$; derivative: $2x$
25. Simplify to $x - 1$; derivative: 1

Part D

26. $4(x+3)^3$
27. $10(2t-1)^4$
28. $6p(p^2+1)^2$
29. $4(6x-1)(3x^2-x)^3$
30. $2(3m^2+1)(m^3+m)$
31. $-12r(1-r^2)^5$
32. $-(x+1)^{-2}$
33. $-4t(t^2+4)^{-3}$
34. $\frac{1}{2}(k+5)^{-1/2} = \frac{1}{2\sqrt{k+5}}$
35. $\frac{n}{\sqrt{n^2+1}}$
36. $2x(x+1)^3 + 3x^2(x+1)^2 = x(x+1)^2(5x+2)$
37. $\frac{(t+1)^2-2t(t+1)}{(t+1)^4} = \frac{1-t}{(t+1)^3}$
38. $\frac{2x+4}{2\sqrt{x(x+4)}} = \frac{x+2}{\sqrt{x(x+4)}}$
39. $\frac{3x^2}{(x+1)^4}$

Part E

40. $\frac{(x^3+1)^5}{5} + C$
41. $\frac{(t^2-3)^6}{6} + C$
42. $\frac{-1}{p^4+2} + C$
43. $2\sqrt{x^3+1} + C$
44. $\frac{(m^2+m)^4}{4} + C$
45. $\frac{(x^2+1)^4}{8} + C$

46. $\frac{(t^2+5)^5}{10} + C$
47. $\sqrt{x^2 + 4} + C$
48. $\frac{(p^3-1)^3}{9} + C$
49. $(x^2 + 1)^5 + C$
50. $(t^3 - 2)^4 + C$
51. $\frac{(3x-2)^5}{15} + C$
52. $\frac{-1}{10(5t+1)^2} + C$

Part F

53. $\frac{2x(x+1)^3 - x^2 \cdot 3(x+1)^2}{(x+1)^6} = \frac{x(2-x)}{(x+1)^4}$
 54. $(x^2 + 1)^{1/2} + \frac{x^2}{\sqrt{x^2+1}} = \frac{2x^2+1}{\sqrt{x^2+1}}$
 55. $\frac{-1}{2(x^2+1)^2} + C$
 56. $\sqrt{x^2 + 9} + C$
 57. $\frac{(x^2+2x)^5}{10} + C$
 58. Derivative: $6x(x^2 + 1)^2$; Integral: $(x^2 + 1)^3 + C$
 59. Differentiation loses information (the constant), integration adds it back arbitrarily.
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